# Chapter 5. Understanding Your Machine

When you first start fleshing out your game concept, you should concentrate on the dream, the player's role, and the target audience. However, a game concept is not complete without a statement about which machine (or machines) the game runs on. All machines have features and performance characteristics—input and output devices, processor speed, storage space—that define the scope of the game. You need to know the strengths and weaknesses of the different types of machines. The differences go far beyond hardware, however. So far as design (rather than programming) is concerned, the most important things you need to know about these machines are related to who buys them and, above all, how they use them. Some genres of games are better suited to one kind of machine than another. The physical positioning and configuration of the device plays an enormous role in determining what games will be suitable for it.

Different machines also tend to imply particular distribution channels (how the player obtains the game) and business models for selling games to your customers (retail, subscription, free-to-play, and so on). For example, games for the Apple iPhone are available only by download from the Apple's App Store, while blockbuster console titles are normally sold in retail stores. We'll address those issues in Chapter 6, "Making Money from Your Game."

There isn't room in this book to go into the exact hardware specifications of the many different kinds of game machines available, and even if there were, the information would be out of date in a few months. Instead, this short chapter will examine three broad categories of game machines: home game consoles, personal computers, and portable devices. We'll end with a look at some more specialized gear.

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## HOME GAME CONSOLES

Use the term *video game* and most people will think of home game consoles, even today when games have become at least as popular on mobile phones as they are on consoles. In this section, we'll look at how people use consoles and how that affects the games that we make for them.

### Typical Use

A home game console is usually set up in the living room or a bedroom. The player sits or stands holding a dedicated controller 3 to 10 feet away from the television that serves as its display. Although modern high-definition digital televisions are a great improvement over the analog sets that the early consoles used, the player is still too far from the screen to see small details or to read fine print conveniently. This means that games designed for the home console machine cannot be as intricate as the personal computer games can. The graphics have to be simpler and bolder.

Because several people see the display at once, and because all consoles allow for at least two controllers, console machines are excellent for multiplayer local games in which all players look at the same screen. They're ideal for party games, dance games, sports games, and any form in which players take turns while others watch, such as *Wii Sports: Bowling*. On the other hand, console games are not so good for games that require individual players to have secret knowledge, because every player can see what every other player is doing on the screen. Also, because they're set up in a private house, players can use them for a long time with (comparatively) few distractions, so games can be designed and paced with this in mind. Mobile games, in contrast, need to be played for minutes, not hours.

Home consoles are dedicated gaming machines. They have no other function as computers; a few efforts were made to make them more general-purpose by adding keyboards—these never really caught on. Although it is possible to surf the web on them, the lack of a keyboard and the poor quality of their pointing devices limits what

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you can do with them. Console machines are aimed at dedicated hobbyists and children (or their parents, who actually buy them)—people who are prepared to spend a great deal of time playing games.

### Input Devices

In a console game, the control method and user interface must be manageable with the controller provided. A mouse can point much more precisely than most console controllers. Still, you are guaranteed that every machine ships with a standardized controller, which means you don't have to do the configuration testing that games for the PC sometimes require. Generally speaking, hardware developers create a much larger variety of input devices for console machines—steering wheels, guitars, conga drums, fishing rods, balance boards, and so on—than they do for the PC. Because the console manufacturers rigorously test any device with their logo on it, you can be confident that these devices are compatible with their machines.

In 2006 Nintendo dramatically challenged existing conventions by introducing the Wii with its revolutionary motion-sensitive controller. Many casual players find the vast array of buttons and joysticks on the traditional game controller daunting. The Wii has made video games intuitive and easy to learn, with the result that Wii machines are being used in all sorts of unexpected ways—as therapy for injured, disabled, and elderly people, for example.

The Kinect motion-capture device for the Microsoft Xbox represents another major step, although so far it has had less impact than the Wii did. It, too, is designed to be used in a room with some dedicated open space for play such as a living room or den. This makes the Kinect an excellent choice for any game requiring a lot of physical activity. (Chapter 12, "Creating the User Experience," discusses the Kinect in much more detail.)

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### Business Considerations

Home consoles tend to have graphics display hardware of comparable power to the graphics hardware in personal computers, but they have slower central processing units and less RAM than personal computers possess. Because consoles sell for $200 to $400 once they have been available for a year or so, the manufacturer has to cut the hardware design to the bone to keep the cost down. This means that, as computing devices, the most expensive console is always less powerful than the most expensive personal computer, and more difficult to program. On the other hand, their low price and ease of maintenance means that far more consoles are in players' hands, which creates a larger market for their games.

The home console differs from the personal computer in another important way. Console manufacturers won't let just anyone make a game for their machine. You have to have a license from the manufacturer and their approval for your game idea—and they tend to be reluctant to approve anything controversial (though what was controversial 20 years ago now seems tame, thanks to shifting public tastes). Once the game is ready, you also have to submit it to the manufacturer for extensive testing before releasing it. Console manufacturers are very anxious that all games played on their systems should be trouble-free and easy to use. They are very protective of their brand's reputation. When developing for personal computers, you aren't constrained this way. You can create any game you want, without anyone's permission, on any subject you like. Obviously some publishers (or distribution channels like the App Store) won't publish games that they feel might be offensive or illegal, and many countries have censorship laws that explicitly prohibit certain content. But the PC is an open platform; you can build games for personal computers without being bound by any contractual limitations.

A related complication is that unlike a PC, the internal workings of a console machine are a trade secret. To get a license to develop for one, you must sign a

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nondisclosure agreement with the manufacturer in which you promise that you will not reveal any of the details to anyone else, a factor that makes it more difficult to get help. In order to build a game for console machines, you must also have a hardware device called a *development kit* or *dev kit*. You must purchase this device from the manufacturer of the console, and it can be extremely expensive. By contrast, anyone who owns a PC can program a PC—it does not require any special hardware to get a program into it.

Note

The Ouya, a so-called microconsole that runs the Android operating system, is an exception to the rule that console development requires an expensive dev kit. Anyone who owns an Ouya can program games for one, and the device is also easy to open and modify. However, the device has received only lukewarm reviews, and it is too early to tell whether it will be a game-changer or a dud. Microsoft supported dev-kit–free development for the Xbox for a few years via their XNA framework, but later abandoned it.

In the last few years, a partial convergence has taken place between the home console and the PC. Consoles now routinely include disk drives that enable the player to store far more data than before, and they all include networking capability as well. Networked play is now commonplace on console machines, and the availability of downloadable games on such services as Xbox LIVE is placing considerable pressure on the traditional retail business model.

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The General-Purpose Set-Top Box: A Marketing Dead End

Over the years various companies have tried to develop a device rather like a console that will serve as a means of delivering gameplay, reference information, multimedia software, and educational products—in effect combining the functions of a game console and a personal computer, along with television-related functions such as video recording. These devices are usually referred to as *set-top boxes,* meaning a box that sits on top of a television set. (This is now an anachronism, as you cannot perch a device on top of a flat-screen TV.) The idea was to provide an inexpensive computer aimed primarily at consuming media rather than doing work.

Two examples were the Phillips CD-I Player and the 3DO Interactive Multiplayer, both of which came out in the early 1990s. Neither was a success in the marketplace. Their marketing campaigns emphasized their utility for reference and multimedia software, but these functions were provided at less cost by the World Wide Web. Eventually the 3DO Company gave up on marketing the Multiplayer as a general-purpose device and concentrated solely on games, but it was too late. In time, they sold off the intellectual property rights to the hardware and simply became a game publisher. They went out of business in 2003.

If anything can be said to have inherited the mantle of the set-top box, it is the Apple iPad. Although inconvenient for typing on, and not really designed for more than one person to use at a time, it is an excellent inexpensive device for gaming and consuming entertainment and educational media—the original vision for set-top boxes.

## PERSONAL COMPUTERS

Personal computers appeared in the marketplace shortly after home game consoles did, and although they seemed to be general-purpose computing devices, the early ones were too underpowered to be that useful for home office or business use. However, they were immediately successful as gaming platforms, and the Commodore 64, Atari ST, Amiga, and other personal computers were wildly popular among gamers and early computer hobbyists, most especially because their owners could develop their own games for them. Magazines devoted to these machines routinely printed source code of games for people to type in, play, and modify. But they didn't achieve acceptance beyond the hobbyist community until IBM introduced the IBM PC in 1981.

Throughout this book you'll see many references to PCs. The IBM PC and its clones, running the Windows operating system, are by far the most popular personal computers for gaming. However, when you see the term *PC,* you shouldn't assume that it means only IBM PC

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clones running Windows, nor desktop machines, since PCs can be laptops or tablet devices. Developers also make games for the Macintosh and for machines running the Linux operating system, and these qualify as PCs too. For our purposes, what really characterizes a PC is the presence of a keyboard and mouse. (Tablet PCs are really portable devices and are covered in that section.)

### Typical Use

A personal computer is likely to be set up away from the communal living space, on a desk. It's designed to be operated by one person while she is sitting. In this case, the player has a keyboard, a mouse, possibly a joystick, and (more rarely) a dedicated game controller such as those for console machines. The player sits 12 to 18 inches away from a high-resolution display. This high resolution means that the game can have subtle, detailed graphics. The mouse allows precision pointing and a more complex user interface. The keyboard enables the player to enter text conveniently.

The player's close proximity to the screen creates an immersive experience. Some players even add extra screens to the left and right of the main screen to display parts of the game world in their peripheral vision; this can help them see opponents better in racing games and in multiplayer battles online. It also contributes to their feeling of immersion in the game world.

The personal computer is quite awkward for more than one person to use at a time. The controls of a PC are all designed for one individual, and even the furniture it usually sits on—a desk—is intended for solitary use. Even laptops, which are more portable, are still inconvenient to use anywhere but on a desk, and that's why they are included here rather than in the later section on portable devices.

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### Input Devices

The keyboard and mouse are unique to the PC and laptop experience. With its 101 keys, the keyboard allows user interfaces to employ many, many buttons, and complex computer role-playing games (RPGs) such as *The Lord of the Rings Online* make heavy use of this feature. The keyboard enables the player to enter text conveniently and send chat messages to other players over a network, something that is much less convenient with console machines.

### Business Considerations

The great advantage of PC development is that anyone can program one; you don't have to get a license from the manufacturer or buy an expensive development station. Consequently, personal computers are at the cutting edge of innovation in computer gaming. They're the platform of choice for interactive art and other experimental forms of interactive entertainment. Mobile devices are similarly easy and inexpensive to develop for, but they haven't achieved the same level of acceptance as platforms for artistic expression that PCs have. This may soon change, however.

The great bane of PC development is that no two machines are alike. Because they're customizable, millions of configurations are possible. In the early days of the game industry, this was a real nightmare for programmers. Fortunately, the Windows and Macintosh operating systems, and middleware solutions like Unity, have solved many of these problems by isolating the programs from the hardware. Still, games tend to require more from the machine than other applications do, and configuration conflicts still occur.

PC games may be divided into two general and quite different categories: stand-alone games, which the player installs on his machine like any other program, and browser-based games that run inside a web browser.

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### Stand-Alone Games

A stand-alone PC game can use the full power of the PC, assuming the player isn't running any other applications at the same time. Of all games played on consumer equipment, stand-alone PC games can be the most visually spectacular. Home game consoles are sold in distinct hardware "generations," and a given manufacturer's console won't change until the next generation. In contrast, personal computers evolve constantly, so game developers can take advantage of the latest hardware innovations. Crytek's *Crysis* is a good example of a game that requires a very powerful machine for a player to enjoy it fully.

This doesn't mean that stand-alone PC games always demand high-end equipment, or that they should; it only means that if you want to develop for the highest-end gear, you should build stand-alone PC games. That choice usually limits the size of your market to the truly dedicated hobbyist gamer who will have purchased the latest PC technology. On the other hand, many stand-alone games are aimed at the middle of the range and do very well.

### Browser-Based Games

Browser-based games are a rapidly growing sector of the game market. They have one huge advantage over stand-alone games: Because they run in a web browser, they are somewhat isolated from the machine's hardware. A browser-based game can run on a Windows PC, Macintosh, or Linux machine with no modifications. This advantage comes at a price, however; browser-based games cannot take full advantage of the machine's capabilities. Most browser-based games—and there are thousands—are 2D games aimed at the casual player. They are often written in Java or Adobe's ActionScript language, which works with Flash Player.

Beginning with browser-based games is an excellent way to get started building small games, because you don't have to know much about the machine's hardware and the game will be more portable between platforms.

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## PORTABLE DEVICES

Portable devices are a hugely popular and inexpensive form of entertainment. They began as dedicated handheld devices that could play only a few built-in games (sometimes only one), and were primarily regarded as toys for children. Since then, they have branched out and out, widening their demographics to adults and becoming moderately powerful and flexible computing devices. Their CPUs are slower than their console counterparts but still have enough power to run sophisticated games. The Sony PSP represented a huge jump in the power and display quality of dedicated handheld game machines.

Mobile phones and other networked devices such as the Nintendo DS have a distinct advantage over traditional game handhelds: They permit portable networked play. Players can compete against others while riding on trains or waiting for an appointment. Setting up a networked game on mobile phones usually requires having a deal for data transmission services with a cellular service provider. Also, unlike dedicated game machines, for the most part, phones do not require a license from the hardware manufacturer. Anyone can write a program for a mobile phone.

### Typical Use

A portable device is designed to be carried around and used by one person. Such devices range in size from quite small mobile phones up to tablet computers that can be nearly the size of a laptop. A key distinction among use cases is whether the device will fit into a pocket. If it does, it can go almost anywhere; if it doesn't, the user has to have a purse, briefcase, or backpack to put it in. A few intermediate-sized devices that will fit only into a large pocket also exist; these are known by the rather unfortunate name *phablet*. The Samsung Galaxy Note series are phablets.

Because the user holds a portable device in her hands, her eyes are about a foot from the screen—closer than with any other device. However, screen size is a critical limitation and unfortunately, it is different on just about every make and model of device. Screen resolution varies

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considerably as well. This means that user interface elements have to be very simple and clear—even simpler than in a console game, which displays on a large TV screen.

Because portable devices are frequently used in public, every game must include a way to turn off the sound, even though the device itself also includes a master volume control. Players sometimes want to leave the master volume for their device on so they can hear notifications of incoming text messages or phone calls, but turn off the game sound so they don't annoy the people around them.

Battery life also affects the typical use of portable devices. Games tend to be CPU and graphics-intensive applications. You cannot expect a player to play for long periods without being able to save the game as you can on other machines, because at any moment he may have to take an incoming phone call or power down the device to save the battery.

#### Input Devices

One peculiarity of portable devices is that the user must support the device's weight while she uses it. Holding it up can sometimes interfere with its usability. Players using console controllers also have to hold them up, but because most console controllers don't have screens in them (the Wii U's GamePad is an exception), a controller can be better designed to fit the hands. Also, the consequences of dropping an ordinary controller are less dire. Portable devices tend to be fragile.

Handheld machines support few add-on features; the input and output devices are usually fixed. These machines have a smaller number of buttons (if any at all, in the case of smartphones) than a console controller does. However, unlike either consoles or PCs, portable devices frequently include global positioning systems, enabling you to create augmented reality games that are played by moving around in the real world. The Nintendo DS and 3DS series are unique among dedicated game handhelds in that they have two screens, one of which is touch-sensitive and can be used with a stylus.

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### Dedicated Game Handhelds

The cheapest handheld machines offer a fixed set of built-in games, but the more versatile handhelds accept games stored on cartridges, and the PSP supports a small optical disk. Cartridges store much less data than the CD-ROMs or DVD discs that home consoles and computers use. Designing for a cartridge places severe limits on the amount of video, audio, graphics, and animation you can include in the game. Because they're solid-state electronics, though, the data on a cartridge is available instantly. There's no delay in loading data, as there is with optical media devices, and little danger that the cartridge will be damaged.

The handheld game market is potentially lucrative, but creating a game for one tests your skills as a designer. With less storage space, you have to rely on gameplay rather than content to provide the entertainment. The normal way people play these games is for a few minutes at a time, not an hour or so, and they might be interrupted at any time; your game must be designed to meet these expectations. As with home console machines, to develop for dedicated handheld game machines, you must have a license from the manufacturer. Games for dedicated game handhelds still tend to be sold at retail, although there is a growing digital distribution option.

### Mobile Phones and Wireless Devices

Mobile phones now have enough computing power to play decent games. The worst thing about developing for a wide range of mobile phones is the lack of standardization. The screens are all different sizes and color depths; the processors are different; the operating systems are different. Even the layout of the buttons is nonstandard, making it difficult to be certain what user interface design is convenient across a range of phones. Expect to do a lot of compatibility testing on different hardware if you develop for mobile phones.

Unlike dedicated game handhelds, mobile phones and tablets sell their games in a number of different ways. Some are simply for sale for a flat fee, usually for far, far

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less money than console games cost. Some are sold as time-based subscriptions. Many are given away free and then paid for through what are called *in-app purchases (IAPs)*—small payments that allow a player to continue to buy items, play past a certain point, or play more frequently. (This is usually known as the *free-to-play* model and is discussed at greater length in Chapter 6.)

As we noted earlier, home game consoles are closed systems, and developing for them normally requires a license from the manufacturer; PCs are open systems that anyone can develop for. Mobile phones are split on this issue. Apple does not require a license to develop a game (or any other application) for their devices, but they do insist that the games be distributed only through their App Store. They also exercise (sometimes rather capricious) editorial control over what content they will permit in games, as well as books, music, and other content that they sell through iTunes. However, this control does not take the form of prior restraint as it sometimes does on consoles. Developers occasionally put a product on the App Store for sale and only later discover that it has been taken down.

Google's Play store for Android devices also exercises editorial control over the content that it permits. However, it is not necessary to distribute your game via the Play store at all. If you want to avoid compliance with Google's policies, your players with Android phones can download the installation file directly from a website, or receive it as an attachment to an e-mail message. This is not ideal from a marketing perspective, because most casual users will look for games on the Play store and nowhere else, and they must have enabled their phone to accept applications from unknown sources.

It has become so easy to develop games for mobile phones that there is a huge glut on the market, and many developers never make a profit. On the other hand, those that are lucky enough to make a hit can earn spectacular amounts of money. *Angry Birds* is one such, but it's worth remembering that Rovio made 51 games that were not so successful first.

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## OTHER DEVICES

Games show up on all sorts of other devices these days. The more specialized the device, the more important it is to understand clearly its technical limitations and its audience.

Airlines are starting to build video games into their seats; these games tend to be aimed at children or the casual market. Video gambling machines, too, enjoy growing popularity. Because they are heavily regulated and not sold to consumers, they really constitute an industry unto themselves, but video gambling games require programmers and artists just like any other computer game. And, of course, arcade machines, although not as popular as they once were, still provide employment to game developers. Finally, there are large location-based entertainment systems such as ride simulators in theme parks. Instead of providing an actual thrill ride, these devices use seats on platforms moved by pistons, combined with a large-screen display, to create an illusion of movement. The customers can participate in the experience with input devices attached to their seats.

Because these devices occupy niche markets and often have peculiar design restrictions, this book doesn't address them in detail.

## SUMMARY

In this chapter we looked at three broad categories of game devices: home game consoles, personal computers, and portable devices. Each has its own strengths and weaknesses, and the convergence among them will never be total. Consoles are best used in the living room, where there is some controlled physical space to play in; PCs are best used on a desk by one person; portable devices trade off game complexity for convenience and ease of development. When you think about what machine to put your game on, bear these characteristics in mind.

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## **DESIGN PRACTICE** EXERCISES

1. Certain genres are more often found on one kind of machine than on another. Write an essay explaining which machine each genre works best on and why. How do the machine's features and the way that it is used in the home or in the hand facilitate or hinder the gameplay in each genre?
2. Visit a retail store and look at the shelves, and then go to an online app store. Note the genres you observe in each place and count the number of games available. What is different about the games or genres available for the different machines?

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